



Semester I Examinations 2025–2026

Course Instance Codes(s) 4BCT1, 4BMS2, 4BS2, 4PHO1, 4PHY1, 3SPS1, 1AL1
Exam(s) 4th Science.
Module Codes MA4102
Module **Algebraic Foundations of Quantum Computing**

Paper No 1

External Examiner Dr. Michael Mackey
Internal Examiner(s) Dr. James Cruickshank
★ Dr. Mark Howard

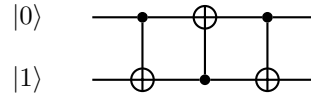
Instructions: **Answer FOUR questions**

Duration 2 Hours
Number of Pages 3 (including this page)
Discipline Mathematics

Requirements:

Release in Exam Venue	Yes ☒	No ☐
Release to Library	Yes ☒	No ☐
Repeat	Yes ☐	No ☒
Statistical/ Log Tables:	State Examination Commission's <i>Formulae and Tables</i>	
Handout	None	
Cambridge Tables	None	
Graph Paper	None	
Log Graph Paper	None	
Other Materials	None	
Graphic material in colour	Yes ☐	No ☒

1. (a) [7 MARKS] Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ be an arbitrary qubit state. By finding γ and δ such that $|\psi\rangle = \gamma|+\rangle + \delta|-\rangle$, determine the probability (as a function of α, β) of obtaining the outcomes "+" and "-" upon measuring $|\psi\rangle$ in the $\{|+\rangle, |-\rangle\}$ basis.
- (b) [6 MARKS] Derive the same results as part (a) using a suitable projective measurement, remarking upon how the definition of a projective measurement is met.
- (c) [6 MARKS] Write out the truth table for a controlled-Hadamard, C_H , gate.
- (d) [6 MARKS] What is the output state for the following circuit



2. Consider the qutrit state given by

$$|\psi\rangle = \frac{1}{\sqrt{14}}(|0\rangle + 2i|1\rangle + 3i|2\rangle)$$

- (a) [7 MARKS] If a measurement is made in the computational basis $\{|0\rangle, |1\rangle, |2\rangle\}$, what is the probability of seeing the outcome "2", and what is the post-measurement state in that case?
- (b) [4 MARKS] Write out the density matrix $\rho = |\psi\rangle\langle\psi|$.
- (c) [6 MARKS] If we apply to $|\psi\rangle$ the unitary

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix}$$

what is the subsequent probability of seeing the outcome "2" in a computational basis measurement?

- (d) [8 MARKS] Prove whether or not the following 2-qubit states are entangled

$$(i) \quad \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix} \quad (ii) \quad \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

3. (a) [13 MARKS] Let $|a\rangle$ be an n -qubit basis state (i.e., $a \in \{0, 1\}^n$). Define $|\phi\rangle$ to be

$$|\phi\rangle = \frac{1}{\sqrt{2^n - 1}} \sum_{x \in \{0, 1\}^n, x \neq a} |x\rangle.$$

In the 2D subspace spanned by $\{|a\rangle, |\phi\rangle\}$, we may write the Grover iterate as a 2×2 matrix G . For searching a list of length four, write out the matrix G , the superposition of all basis states $|\psi\rangle$, and finally $G|\psi\rangle$ (all in the $\{|a\rangle, |\phi\rangle\}$ basis).

- (b) [12 MARKS] Calculate the Quantum Fourier Transform of the following qudit states

$$(i) \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ i \\ 0 \end{pmatrix} \quad (ii) \quad \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ e^{\frac{2\pi i}{3}} \\ e^{-\frac{2\pi i}{3}} \end{pmatrix}$$

4. (a) [7 MARKS] Derive the state $|\psi^\perp\rangle$ orthogonal to the state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Hence find the unitary U mapping $\{|0\rangle, |1\rangle\}$ to $\{|\psi\rangle, |\psi^\perp\rangle\}$ and confirm that $UU^\dagger = \mathbb{I}$.
- (b) [6 MARKS] Prove the quantum no-cloning theorem i.e., the fact there does not exist a 2-qubit unitary that maps

$$|\psi\rangle |0\rangle \mapsto |\psi\rangle |\psi\rangle$$

for every qubit state $|\psi\rangle$.

- (c) [6 MARKS] Simplify the following

$$(\langle 0| \otimes \mathbb{I})(\alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle).$$

- (d) [6 MARKS] Let a set of Kraus operators be $\{E_k\}_{k=0}^3 = \frac{1}{2}\{\sqrt{4-3\lambda}\mathbb{I}, \sqrt{\lambda}X, \sqrt{\lambda}Y, \sqrt{\lambda}Z\}$ with $\{\mathbb{I}, X, Y, Z\}$ denoting the usual identity and Pauli operators. The parameter λ is in the range $0 \leq \lambda \leq 1$. Confirm that these Kraus operators make up a trace-preserving channel in the usual way, and describe the overall effect of this channel on an arbitrary state ρ .

5. Let $U_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ and $|\psi\rangle = U_\theta |0\rangle$ and $|\psi^\perp\rangle = U_\theta |1\rangle$

- (a) [8 MARKS] Show that $|\beta_{0,0}\rangle = \frac{1}{\sqrt{2}}(|\psi, \psi\rangle + |\psi^\perp, \psi^\perp\rangle)$ where as usual $|\beta_{0,0}\rangle := \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.
- (b) [9 MARKS] Suppose Alice and Bob start with the state $|\beta_{0,0}\rangle$. Alice applies U_θ^{-1} on her qubit and then measures in the computational basis. What pure state does Bob have if her measurement is zero? What pure state does Bob have if her measurement is one?
- (c) [8 MARKS] Suppose that something called a “parity measurement” on two qubits comprises two projectors $\Pi_{\text{even}} = |00\rangle\langle 00| + |11\rangle\langle 11|$ and $\Pi_{\text{odd}} = |01\rangle\langle 01| + |10\rangle\langle 10|$. Show that the parity measurement obeys the requirements of a projective measurement, and calculate the probability of getting the outcome “even” if the state $|\beta_{0,0}\rangle$ is measured.